N+ - Network security, Maintenance and Troubleshooting procedures

❖ Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

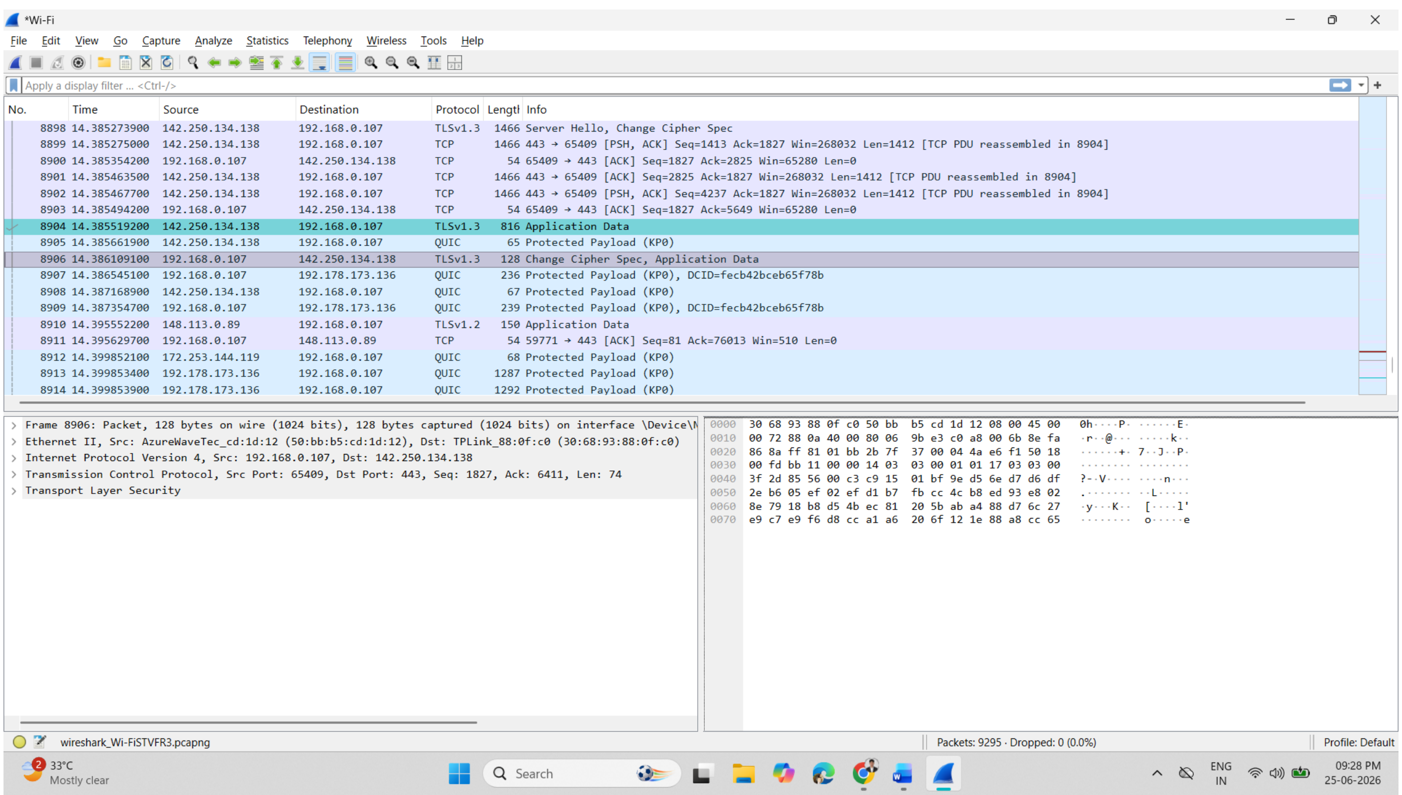
ANS :

- -After installing Wireshark and capturing network traffic while browsing Instagram/YouTube for 2 minutes, the following protocols were commonly observed:

  - TCP (Transmission Control Protocol) – used for reliable data transfer
  - DNS (Domain Name System) – used to resolve domain names into IP addresses
  - HTTPS (Hypertext Transfer Protocol Secure) – used for secure web browsing and streaming

- **Top 3 protocols observed:**

  - TCP
  - DNS
  - HTTPS



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2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

ANS :

- **Ping www.spotify.com (Slower)**

```
C:\Users\HET>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:19::810] with 32 bytes of data:
Reply from 2a04:4e42:19::810: time=20ms
Reply from 2a04:4e42:19::810: time=18ms
Reply from 2a04:4e42:19::810: time=17ms
Reply from 2a04:4e42:19::810: time=17ms

Ping statistics for 2a04:4e42:19::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 20ms, Average = 18ms
```

- **Ping www.zomato.com (Faster)**

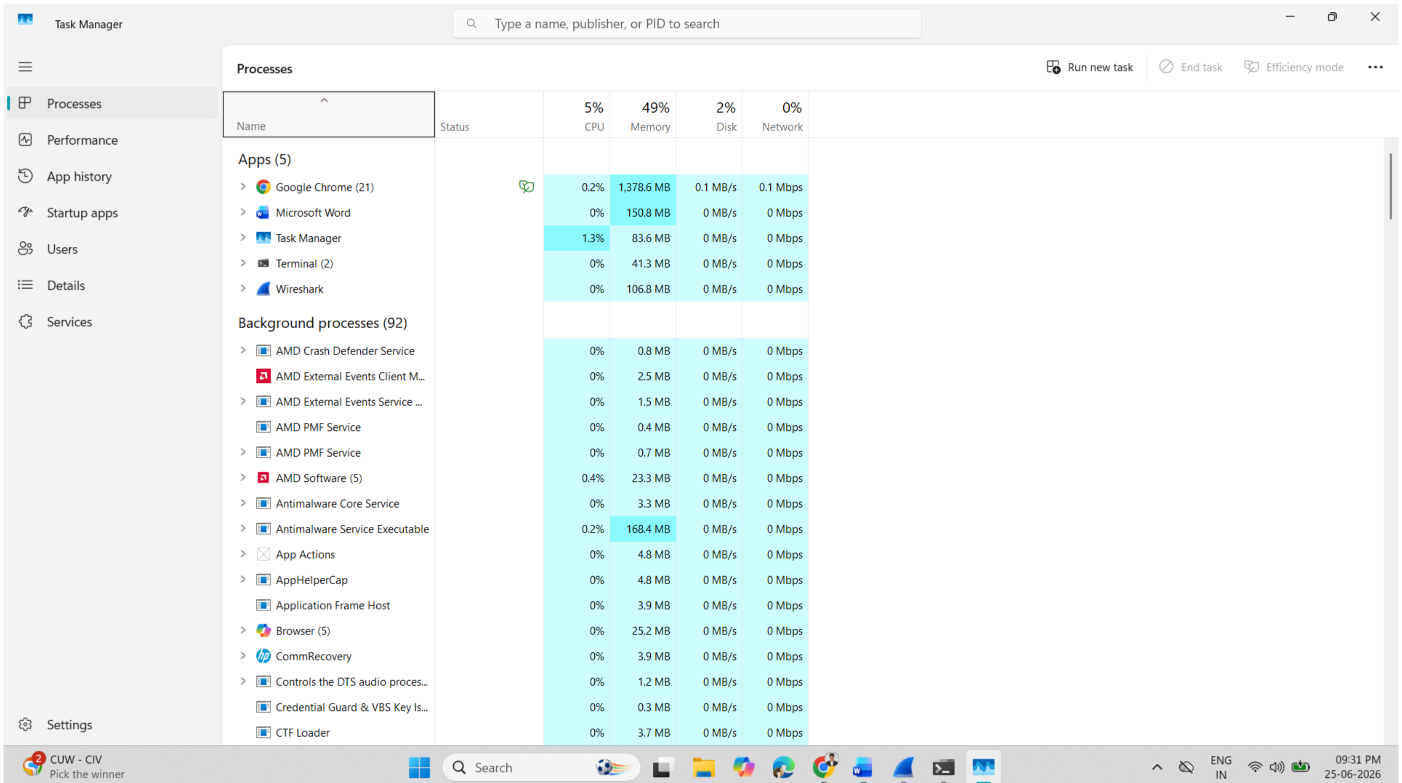
```
C:\Users\HET>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:4::17da:f99a] with 32 bytes of data:
Reply from 2600:140f:4::17da:f99a: time=23ms
Reply from 2600:140f:4::17da:f99a: time=17ms
Reply from 2600:140f:4::17da:f99a: time=15ms
Reply from 2600:140f:4::17da:f99a: time=12ms

Ping statistics for 2600:140f:4::17da:f99a:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 23ms, Average = 16ms
```

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

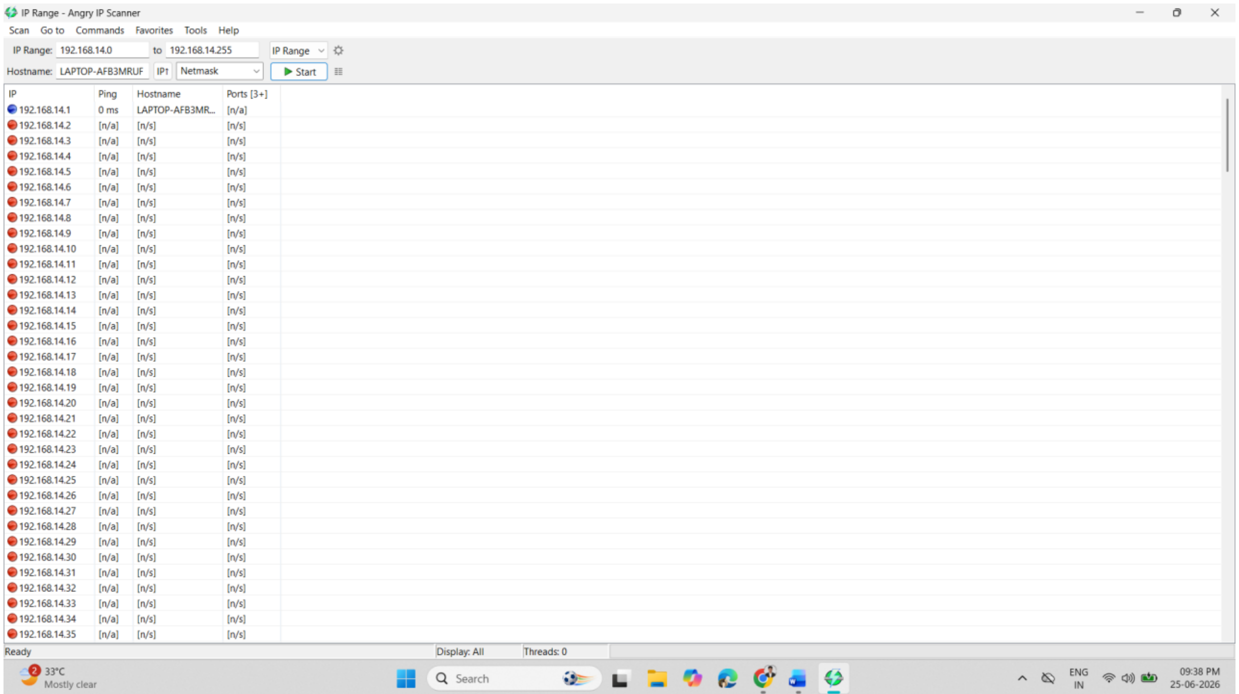
ANS :



Process Name	Application	Memory Usage
chrome.exe	Google Chrome	800 MB – 1500 MB

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint : Make sure your firewall allows scanning, and only scan your own home/college network**

ANS :



- **192.168.14.1** : LAPTOP-AFB3MRUF

5. **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

ANS:

- **Cable Tester** – Used to check whether Ethernet cables are properly connected or damaged.
- **Crimping Tool** – Used to attach RJ45 connectors to network cables.
- **Multimeter** – Used to measure voltage, continuity, and detect electrical issues in network hardware.